

NUMERICALS ON CT

Q1: A current transformer has single-turn primary and a 100- turn secondary winding. The secondary winding of purely resistive burden of 1.5 Ω draws a current of 6 A. The magnetising ampere-turns is 60 A. Supply frequency is 50 Hz and core cross-sectional area is 800 mm² . Calculate the ratio and phase angle of the CT. Also find the flux density in the core. Neglect flux leakage, iron losses and copper losses.

Solution:

Neglecting copper losses and flux leakage implies that secondary winding resistance and reactance are not be considered. The burden of 1.5 Ω thus can be assumed to be the burden of the entire secondary circuit.

Secondary induced voltage $E_s = \text{Secondary current} \times \text{Secondary burden} = 6 \times 1.5 = 9 \text{ V}$

As the secondary burden is purely resistive, the secondary current is in phase with the secondary induced voltage, i.e., the phase angle difference between E_s and I_s , $\delta = 0$, and power factor is unity.

Given that iron losses, and hence the loss component of current can be neglected, i.e., $I_C = 0$.

No-load current in this case is thus simply equal to the magnetising component of current, i.e., $I_0 = I_M$

Now, magnetising component of no-load current is given by

$$I_M = \frac{\text{Magnetising mmf}}{\text{Primary winding turns}} = \frac{60}{1} = 60 \text{ A}$$

Secondary winding current $I_s = 6 \text{ A}$

Turns ratio = $n = 100/1 = 100$

Reflected secondary winding current = $nI_s = 100 \times 6 = 600 \text{ A}$

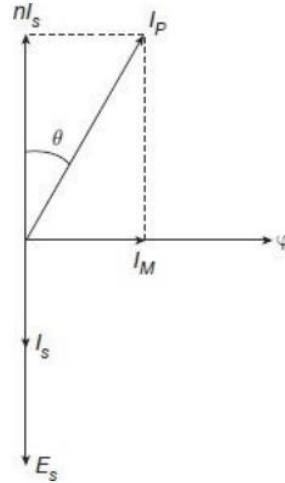
Referring to the phasor diagram shown in the figure, the primary current can be calculated as

$$I_p = \sqrt{(nI_s)^2 + (I_M)^2} = \sqrt{600^2 + 60^2} = 603 \text{ A}$$

$\therefore$ actual transformation ratio

$$R = \frac{I_p}{I_s} = \frac{603}{6} = 100.5$$

$$\text{Phase angle} = \theta = \tan^{-1} \left(\frac{I_M}{nI_s} \right) = \tan^{-1} \frac{60}{600} = 5.7^\circ$$



From the relation $E_S = 4.44f\phi_m N_S$, the maximum flux ϕ_M in the core can be calculated as

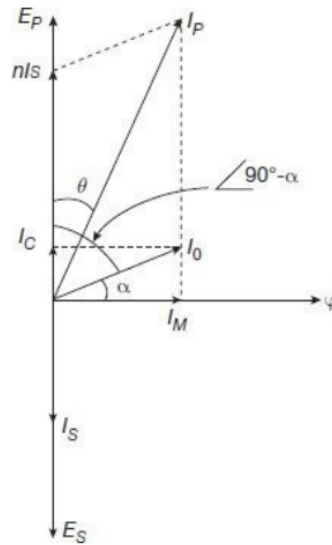
$$\phi_m = \frac{E_S}{4.44fN_S} = \frac{9}{4.44 \times 50 \times 100} = 0.41 \times 10^{-3} \text{ Wb}$$

Given, area of core = $800 \text{ mm}^2 = 800 \times 10^{-6} \text{ m}^2$

$$\therefore \text{maximum flux density } B_m = \frac{0.41 \times 10^{-3}}{800 \times 10^{-6}} = 0.51 \text{ Wb/m}^2$$

Q2: A ring core type CT has a ratio of 2000/10. When operating at rated primary current with a secondary burden of noninductive resistance value of 2Ω , takes a no-load current of 2 A at power factor of 0.3. Calculate (i) the phase angle difference between primary and secondary currents, and (ii) the ratio error at full load.

Solution:



The burden being purely resistive, the secondary current I_S and secondary induced voltage E_S will be in the same phase as shown in the figure. Therefore, secondary winding power factor will be unity and the phase angle difference between I_S and E_S , $\delta = 0$.

Give, no-load current $I_0 = 2$ A, at power factor of 0.3.

$$\therefore \cos (90^\circ - \alpha) = 0.3 \text{ or } \alpha = 17.46^\circ$$

$$\text{Nominal ratio } K_n = \frac{2000}{10} = 200$$

Without any turns compensation, nominal ratio equals the turns ratio, thus $n = K_n$

Under full load rated condition, primary current = 2000 A and secondary current = 10 A.

$\therefore$ transformation ratio

$$R = n + \frac{I_0}{I_S} \sin (\delta + \alpha) = 200 + \frac{2}{10} \sin (0^\circ + 17.46^\circ) = 200.06$$

$$\therefore \text{ratio error} = \frac{\text{Nominal Ratio} - \text{Actual Ratio}}{\text{Actual Ratio}} \times 100\% = \frac{K_n - R}{R} \times 100\%$$